Intelligent Resume Parsing & Skill Proficiency Analysis

R26-IT-082

Project Proposal Report

Jayantha CH – IT22226914

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

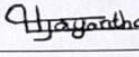
Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Jayantha CH	IT22226914	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Mr. Dhammika De Silva	13/03/2026	

ABSTRACT

The increasing use of online recruitment platforms has resulted in organizations receiving large volumes of job applications. To manage this process, many organizations rely on automated tools such as Applicant Tracking Systems (ATS) for resume screening. However, most existing systems depend on keyword-based matching techniques that only detect the presence of skills without evaluating their actual depth or relevance. This limitation may lead to inaccurate candidate evaluation and the potential rejection of qualified applicants.

To address this issue, this research proposes an Intelligent Resume Parsing & Skill Proficiency Analysis module that uses Natural Language Processing (NLP) techniques to extract and analyze information from unstructured resumes. The system identifies key entities such as skills, education, and work experience, and introduces a context-aware skill proficiency estimation approach that evaluates candidate skills based on contextual factors such as experience duration and project descriptions.

The system will be developed using Python, with NLP processing implemented using SpaCy, CRF-based Named Entity Recognition (NER) through sklearn-crfsuite, and resume text extraction using pdfplumber or PyMuPDF. Extracted candidate information will be stored in a MongoDB database.

The proposed solution aims to improve the accuracy and efficiency of resume screening while supporting data-driven recruitment decisions. This system can benefit organizations by reducing manual screening effort and enhancing the effectiveness of candidate selection.

Keywords: Resume Parsing, Natural Language Processing (NLP), Skill Proficiency Analysis, AI Recruitment, Named Entity Recognition (NER).

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project supervisor Mr. Dhammika De Silva for his valuable guidance, continuous support, and constructive feedback throughout the preparation of this research proposal. His insights and encouragement greatly contributed to improving the quality and direction of this research work.

I would also like to extend my appreciation to my co-supervisor Prof. Samantha Rajapakshe for his guidance, expert advice, and support during the development of this project.

Furthermore, I would like to thank my fellow group members and colleagues for their cooperation and valuable discussions during the planning of this project.

TABLE OF CONTENT

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 7
LIST OF FIGURES 8
LIST OF ABBREVIATIONS 9
LIST OF APPENDICES 10
1. INTRODUCTION 11
1.1 Background and Context 11
1.2 Literature Review 12
1.3 Research Gap 15
2. OBJECTIVE 16
2.1 Main Objective 16
2.2 Specific Objectives 16
3. METHODOLOGY 17
3.1 Research Approach 17
3.2 Technical Methods and Algorithms 17
3.3 Tools and Platforms with Justification 17
3.4 Data Collection and Ethical Considerations 18
3.5 Validation Strategy and Performance Metrics 18
4. HIGH-LEVEL SYSTEM ARCHITECTURE 19
4.1 Overview of the Proposed Solution and Its Components 19
4.2 System Diagram Individual Components 19
4.4 Component Interactions 20
4.5 Data Flow Explanation 20
4.6 Scalability and Security Considerations 20
5. USER REQUIREMENTS 21
5.1 Stakeholder Identification 21
5.2 Functional Requirements 21
5.3 Non-Functional Requirements 22

6. COMMERCIALIZATION PLAN 23
6.1 Target Market and Customer Persona 23
6.2 Value Proposition 23
6.3 Revenue Model 23
6.4 Cost Estimation 23
6.6 Competitive Advantage 24
6.7 Intellectual Property Considerations 24
7. BUDGET AND JUSTIFICATION 25
7.1 Software Licenses 25
7.2 Cloud Services 25
7.3 Human Effort Cost 25
7.4 Total Estimated Budget 26
7.5 Budget Justification 26
8. WORK BREAKDOWN STRUCTURE (WBS) 27
9. GANTT CHART 28
10. REFERENCES LIST 29
11. APPENDICES 30

LIST OF TABLES

Table 1.Software Tools and License Cost Estimation.....25
Table 2.Cloud Services Cost Estimation.....25
Table 3.Human Effort Cost Estimation.....26
Table 4.Total Estimated Project Budget.....26
Table 5.Risk Analysis and Mitigation Strategies 32

LIST OF FIGURES

Fig. 1.Resume Analysis System Architecture 13
Fig. 2. Block Diagram of Job Parsing Module 14
Fig. 3.AI-Driven Intelligent Talent Acquisition System Architecture 19
Fig. 4.Intelligent Resume Parsing & Skill Proficiency Analysis System Architecture 19
Fig. 5.Work Breakdown Structure 27
Fig. 6.Gantt Chart 28
Fig. 7.Resume Upload Interface 31
Fig. 8.Candidate Skill Profile Dashboard 31
Fig. 9.Candidate Ranking & Analysis Page 32

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
DL	Deep Learning
NER	Named Entity Recognition
CRF	Conditional Random Field
ATS	Applicant Tracking System
HR	Human Resources
PDF	Portable Document Format
DB	Database
UI	User Interface
SDG	Sustainable Development Goal

LIST OF APPENDICES

Appendix A:Survey Instruments.....30
Appendix B:Preliminary UI Mockups.....30
Appendix C:Risk Analysis.....32
Appendix D:Ethical Documents.....33

1. INTRODUCTION

1.1 Background and Context

The rapid growth of online recruitment platforms and Applicant Tracking Systems (ATS) has significantly transformed the hiring process in modern organizations. Companies now receive a large number of job applications for each vacancy, making manual resume screening difficult and time-consuming. To manage this challenge, many organizations rely on automated resume screening and parsing systems. However, most existing systems primarily depend on keyword-based matching techniques, which often fail to accurately represent a candidate's actual skills and experience [1]. As a result, qualified candidates may be rejected during the initial screening stage while less suitable candidates may pass through the filtering process.

This issue affects several stakeholders including job seekers, recruiters, and organizations. Job seekers may lose employment opportunities because automated systems cannot fully understand the contextual meaning of their work experience, project descriptions, or skill usage. Recruiters and human resource professionals must review a large number of resumes within a limited time, which increases the likelihood of biased or inconsistent evaluation. Organizations are also negatively affected by poor hiring decisions, which may result in lower productivity, increased recruitment costs, and employee turnover [2].

Statistical evidence highlights the scale of this challenge. Studies show that recruiters typically spend only a few seconds reviewing a resume during the initial screening process, and large organizations often receive hundreds or thousands of applications for a single job position [3]. Because of this high volume, many companies depend on ATS platforms to automate resume filtering. However, most of these systems still rely on simple keyword extraction techniques that do not evaluate the depth, duration, or relevance of candidate skills. Consequently, important contextual information such as project experience, role responsibilities, and real-world skill application is often ignored.

Existing resume parsing tools mainly focus on extracting structured information such as education, work experience, and skill keywords from resumes. While these systems improve the organization of applicant data, they have several limitations. Most systems treat skills as simple binary attributes, meaning they only detect whether a skill appears in the resume without evaluating the candidate's level of expertise or practical experience. In addition, traditional systems rarely consider contextual factors such as experience duration, project complexity, or role relevance, which are important indicators of actual skill proficiency [4].

To address these limitations, this research proposes an Intelligent Resume Parsing and Skill Proficiency Analysis module using Natural Language Processing (NLP). The system will extract structured information from resumes and analyze contextual factors such as experience duration,

project descriptions, and role relevance to estimate candidate skill proficiency. Unlike traditional keyword-based methods, this approach evaluates the depth and relevance of skills, providing a more accurate representation of candidate capabilities and supporting better recruitment decisions.

The proposed research aligns with United Nations Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth, which promotes productive employment and efficient labor market systems [5]. Furthermore, this research aligns with the CoEAI – Centre of Excellence for Artificial Intelligence (CoEAI) research cluster, as it applies Artificial Intelligence techniques such as Natural Language Processing and machine learning to develop intelligent recruitment solutions that address real-world hiring challenges.

1.2 Literature Review

Recent studies highlight the increasing use of Artificial Intelligence (AI) and Natural Language Processing (NLP) to automate recruitment processes and improve resume analysis. Traditional recruitment systems rely heavily on manual resume screening, which is time-consuming and inefficient when organizations receive large numbers of applications. Therefore, many researchers have proposed intelligent systems to automate resume parsing, candidate profiling, and skill matching. Mohammad Fadzir and Hashim [1] proposed SmartResume, an NLP-based resume parser designed to extract structured information such as skills, education, and work experience from resumes. The system uses Named Entity Recognition (NER) implemented through the SpaCy NLP framework, achieving an extraction accuracy of 92.4%. The study demonstrates that NLP techniques significantly improve information extraction efficiency compared to manual screening. However, the system mainly focuses on extracting resume information and does not evaluate skill proficiency or experience depth, limiting its ability to assess real candidate capabilities.

Similarly, Al-Aswad et al. [2] developed an Intelligent Resume Analysis and Job Recommendation System using NLP techniques such as NER and rule-based decision logic. The system extracts structured data from resumes and performs skill gap analysis to recommend relevant jobs and training opportunities. Experimental results showed high accuracy in skill extraction and job recommendation tasks. Despite these strengths, the system primarily focuses on job recommendation and skill gap identification rather than analyzing the contextual depth of candidate experience, which is essential for accurate skill proficiency evaluation.

Kruthika et al. [3] introduced an Automated Resume Parsing and Candidate Profiling System that utilizes NLP++, zoning techniques, and hierarchical parsing to analyze resume sections. The system stores extracted data in a Knowledge Base (KB) and converts it into XML for querying and candidate matching. This approach improves scalability and enables efficient processing of large resume

datasets. However, the rule-based parsing method may struggle with complex sentence structures and domain-specific terminology, which can reduce parsing accuracy in diverse real-world resumes.

Thangaramya et al. [4] proposed an Automated Resume Parsing and Ranking System using Deep Learning (DL) and NLP techniques. The system applies NER and heuristic scoring to rank candidates based on extracted information such as skills, education, and experience. The system achieved approximately 93% information extraction accuracy, demonstrating the effectiveness of machine learning in recruitment automation. However, the ranking mechanism relies mainly on predefined criteria and does not fully capture contextual relationships between skills and professional experience.

Raja et al. [5] presented HireSmart AI, an AI-driven recruitment system designed to automate candidate screening and assessment. The study highlights the widespread adoption of Applicant Tracking Systems (ATS) in modern recruitment, noting that over 98% of Fortune 500 companies use ATS tools. Although these systems improve efficiency, they often suffer from low parsing accuracy and limited contextual understanding, particularly when analyzing resumes with diverse formats and terminology.

Julian and Haripriya [6] proposed an NLP-based Resume Analysis and Adaptive Skill Assessment System that integrates Word2Vec embeddings and machine learning algorithms to categorize job descriptions and match candidates with relevant job roles. The system also generates personalized interview questions and job recommendations. While the approach demonstrates effective automation and personalization, it focuses more on job matching rather than evaluating actual skill proficiency levels derived from resume content.

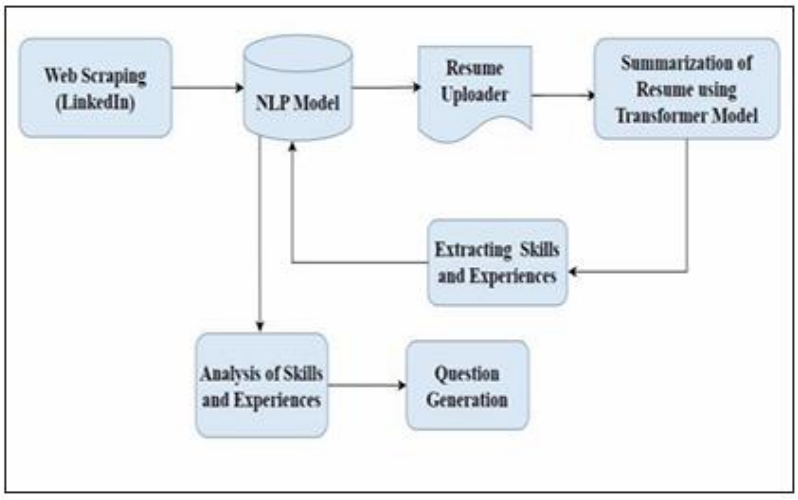


Fig. 1.Resume Analysis System Architecture

The architecture shown in Figure 1 illustrates the main components of an NLP-based resume analysis system including skill extraction, machine learning modeling, and job recommendation modules.

Gangoda et al. [7] introduced an AI-Based Skill Analysis and Skill Matching System designed to address workforce shortages through automated resume analysis and skill alignment. The system applies NLP preprocessing techniques such as tokenization, lemmatization, and cosine similarity to match candidate skills with job requirements. Although this system improves candidate-job matching efficiency, it mainly depends on keyword filtering and similarity matching, which may overlook contextual information related to real skill expertise.

Deepa et al. [8] conducted a Survey on Resume Parsing Models that examines various NLP and machine learning approaches for automated resume analysis. The survey identifies common techniques such as entity recognition, multi-label classification, and deep learning models for resume parsing. Despite these advancements, the study highlights key limitations including difficulty handling diverse resume formats, inability to understand non-standard language usage, and insufficient contextual analysis of candidate skills.

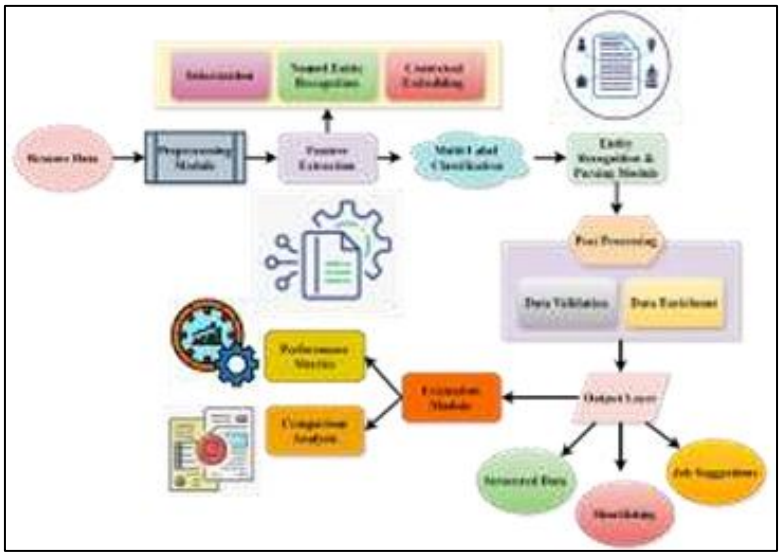


Fig. 2. Block Diagram of Job Parsing Module

Figure 2 presents the general pipeline used in resume parsing systems including preprocessing, feature extraction, entity recognition, and candidate evaluation stages.

In addition to academic research, several commercial recruitment systems such as Applicant Tracking Systems (ATS) provide automated resume screening and candidate ranking features. These platforms primarily focus on keyword matching, rule-based filtering, and skill extraction, enabling organizations to process large volumes of applications efficiently. However, such systems typically lack the ability

to evaluate skill proficiency, contextual experience, and real-world project relevance, which can result in inaccurate candidate evaluation.

Overall, existing research demonstrates significant progress in AI-driven resume parsing and recruitment automation. Techniques such as NER, DL, ML, and NLP-based information extraction have improved resume processing efficiency and candidate matching. Nevertheless, most existing systems emphasize information extraction and keyword-based matching rather than analyzing the contextual depth of candidate skills and experience. This limitation creates an important research opportunity to develop systems that estimate skill proficiency using contextual indicators such as experience duration, project involvement, and role relevance, which forms the foundation of the proposed research.

1.3 Research Gap

Although recent studies have introduced various AI-based recruitment systems, several limitations remain in current resume analysis approaches. Most existing systems focus on resume parsing, keyword extraction, and job–candidate matching using technologies such as Natural Language Processing (NLP), Machine Learning (ML), and Deep Learning (DL). These systems can extract structured information such as skills, education, and work experience using techniques like Named Entity Recognition (NER) [1]–[8]. However, they mainly identify the presence of skills rather than evaluating the actual proficiency level of those skills.

A major limitation identified in existing systems is the reliance on keyword-based skill extraction, commonly used in Applicant Tracking Systems (ATS). These systems treat skills as simple binary attributes, assuming that the presence of a keyword indicates sufficient expertise. This approach ignores important contextual factors such as experience duration, project involvement, and role responsibilities, which are essential for determining real skill proficiency. As a result, candidates with limited experience may be evaluated similarly to those with extensive practical expertise.

Furthermore, most NLP-based resume parsing models emphasize information extraction accuracy rather than analyzing how skills are applied in real-world contexts. Many systems also fail to consider the relationship between skills, projects, and job roles, making it difficult to evaluate the true relevance and depth of candidate capabilities.

Therefore, there is a need for a more advanced resume analysis approach that goes beyond traditional keyword-based methods. This research addresses this gap by introducing a context-aware skill proficiency estimation mechanism within an NLP-based resume parsing system. The proposed approach analyzes contextual factors such as experience duration, project descriptions, and role relevance to estimate actual skill proficiency. This contribution aims to improve the accuracy and reliability of AI-driven recruitment systems and support more effective hiring decisions.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop an NLP-based intelligent resume parsing and skill proficiency analysis module that can automatically extract candidate information from unstructured resumes and evaluate the relevance and depth of candidate skills. The proposed system will utilize Natural Language Processing (NLP) techniques to identify key entities such as skills, education, and work experience from resume documents. In addition to information extraction, the system will introduce a context-aware skill proficiency estimation mechanism that analyzes contextual factors such as experience duration, project descriptions, and role responsibilities. This approach aims to move beyond traditional keyword-based resume screening methods used in Applicant Tracking Systems (ATS) and provide a more accurate evaluation of candidate capabilities to support AI-driven recruitment decision-making.

2.2 Specific Objectives

This research contains the following specific objectives.

- To collect and preprocess resume datasets containing candidate skills, education, and work experience for model development.
- To design and implement an NLP-based resume parsing module to extract structured information from unstructured resume documents.
- To apply NLP techniques such as tokenization and Named Entity Recognition (NER) to identify skills, education, and professional experience.
- To analyze contextual relationships between skills and experience by examining factors such as experience duration, project descriptions, and role responsibilities.
- To develop a context-aware skill proficiency estimation mechanism that evaluates the depth and relevance of candidate skills.
- To evaluate the performance of the proposed system using metrics such as extraction accuracy and skill relevance detection.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach to design, develop, and evaluate an intelligent resume analysis system. The study focuses on creating an NLP-based resume parsing module capable of extracting structured candidate information and estimating skill proficiency using contextual analysis. The research involves designing an artifact (the resume parsing and skill analysis system), implementing the proposed solution, and evaluating its effectiveness using experimental validation. The approach is suitable for developing AI-driven decision-support systems that address real-world recruitment challenges.

3.2 Technical Methods and Algorithms

The proposed system uses Natural Language Processing (NLP) techniques to analyze unstructured resume documents and extract candidate information. The system follows a multi-stage processing pipeline.

First, resumes in PDF format are converted into machine-readable text using pdfplumber or PyMuPDF. After text extraction, preprocessing is performed using SpaCy, including tokenization, text normalization, and stop-word removal to prepare the text for analysis.

Next, Named Entity Recognition (NER) is applied to identify entities such as skills, education, work experience, and project information. The NER model is implemented using a Conditional Random Field (CRF) algorithm through the sklearn-crfsuite library, which is effective for sequence labeling tasks.

The extracted entities are then processed by a context-aware skill analysis module, which evaluates contextual information such as experience duration, project descriptions, and job roles to estimate the skill proficiency level of the candidate.

Finally, the system generates a structured candidate skill profile, summarizing extracted information and estimated skill proficiency to support recruitment decision-making.

3.3 Tools and Platforms with Justification

The proposed system will be developed using the following tools and technologies:

- Python - Python will be used as the main programming language due to its strong support for AI, NLP, and data processing applications.
- SpaCy (NLP Library) - SpaCy will be used for text preprocessing, tokenization, linguistic analysis, and feature extraction from resume text.
- CRF NER (sklearn-crfsuite) - The Conditional Random Field (CRF) model will be used for Named Entity Recognition (NER) to identify key entities such as skills, education, and work

experience from resumes. CRF models are widely used for sequence labeling tasks in NLP and provide high accuracy for structured information extraction.

- pdfplumber / PyMuPDF - These libraries will be used for extracting text from PDF resumes while preserving document layout and formatting information.
- MongoDB - will be used as the NoSQL database to store extracted candidate information and structured skill profiles. It is suitable for this system because it efficiently handles semi-structured data like resume information and allows flexible storage without requiring a fixed schema

These tools are widely adopted in AI and NLP-based text processing systems, making them suitable for developing an efficient resume parsing and skill analysis module.

3.4 Data Collection and Ethical Considerations

The dataset used in this research will consist of anonymized resume data and job-related information collected from publicly available datasets and online recruitment platforms. Additional synthetic resumes may also be generated to simulate different candidate profiles and skill combinations.

During the data collection process, ethical considerations and data privacy regulations will be strictly followed. Personal identifiable information such as names, contact details, and addresses will be anonymized or removed before analysis. The study will ensure that the dataset is used only for research purposes and does not violate candidate privacy.

3.5 Validation Strategy and Performance Metrics

The effectiveness of the proposed system will be evaluated using experimental validation methods. The evaluation will focus on measuring both the accuracy of information extraction and the effectiveness of skill proficiency estimation.

The following performance metrics will be used:

• Extraction Accuracy – Measures how accurately the system extracts skills, education, and work experience from resumes.

• Precision and Recall – Evaluates the correctness and completeness of the extracted information.

• F1-Score – Provides a balanced evaluation of precision and recall in entity extraction tasks.

• Skill Relevance Accuracy – Measures how effectively the system evaluates the contextual relevance of candidate skills.

• System Efficiency – Evaluates the processing time and scalability of the resume parsing system.

These metrics will help determine the effectiveness of the proposed AI-based resume analysis module in improving recruitment decision support compared to traditional ATS-based keyword matching systems.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed system is an AI-Powered Intelligent Talent Acquisition System designed to support recruitment decision-making using Artificial Intelligence (AI) and Natural Language Processing (NLP). The system integrates multiple intelligent modules that analyze candidate data and provide insights for recruiters. The architecture includes four main components: Intelligent Resume Parsing & Skill Proficiency Analysis, Job–Candidate Matching & Explainable AI, AI-Based Interview Evaluation, and Hiring Risk & Early Attrition Prediction.

The Resume Parsing & Skill Proficiency Analysis module, which is the focus of this research, acts as the foundation of the system by converting unstructured resume documents into structured candidate profiles. These structured profiles are then used by other modules for candidate matching, interview evaluation, and recruitment risk analysis.

4.2 System Diagram Individual Components

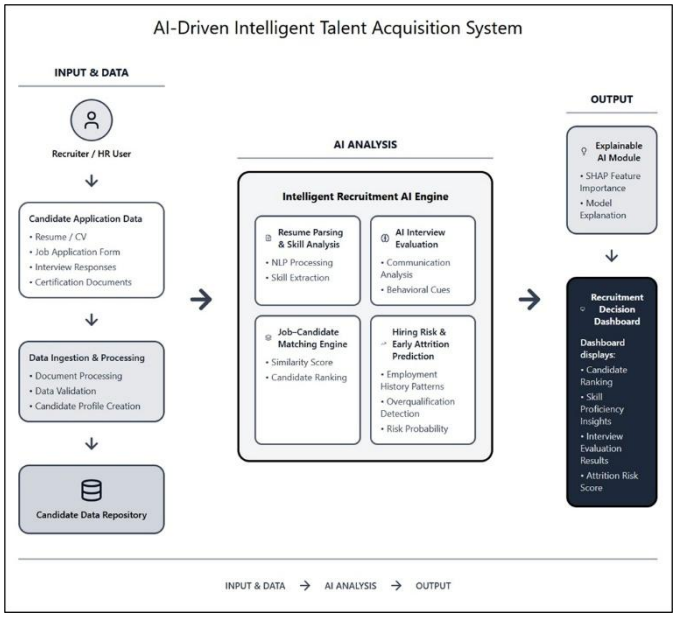


Fig. 3.AI-Driven Intelligent Talent Acquisition System Architecture

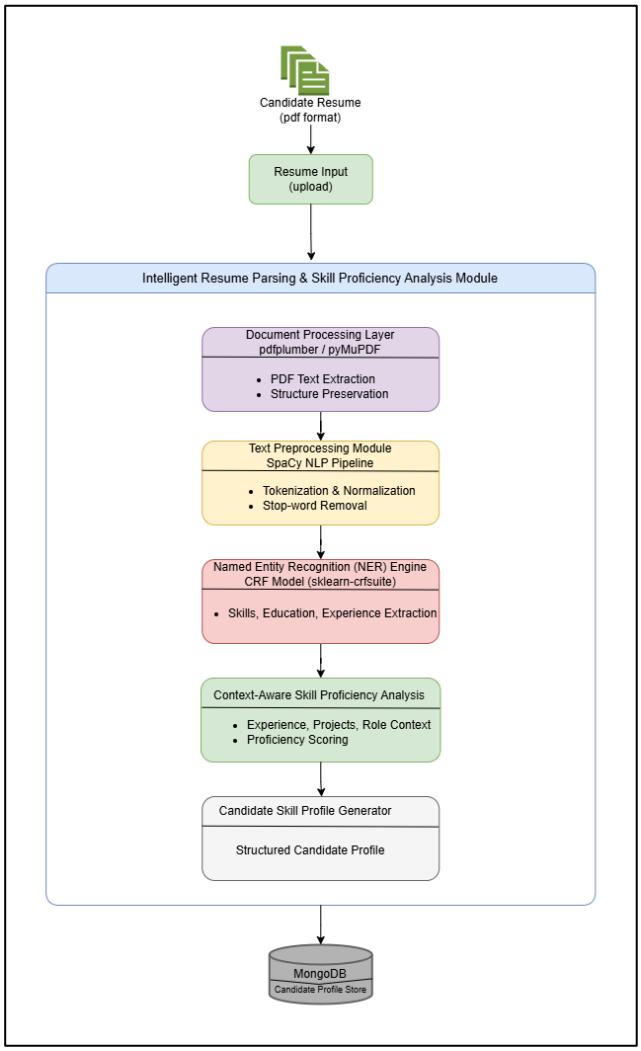


Fig. 4.Intelligent Resume Parsing & Skill Proficiency Analysis System Architecture

Figure 4 illustrate the Intelligent Resume Parsing & Skill Proficiency Analysis module transforms unstructured resume documents into structured candidate information. First, resume files are processed using pdfplumber or PyMuPDF to extract textual content. The extracted text then undergoes NLP preprocessing using SpaCy, including tokenization and text normalization.

Next, a Named Entity Recognition (NER) model based on Conditional Random Fields (CRF) identifies entities such as skills, education, work experience, and projects. These entities are analyzed by a context-aware skill analysis module, which evaluates skill relevance using contextual information such as experience duration and project descriptions. Finally, the system generates a structured candidate skill profile, which is stored in MongoDB for use by other modules.

4.4 Component Interactions

The Resume Parsing module interacts with other components through structured data exchange. After generating candidate profiles, the information is stored in MongoDB, where it can be accessed by other modules. The Job–Candidate Matching module uses these profiles to compare candidate skills with job requirements, while the Interview Evaluation and Attrition Prediction modules use candidate attributes to perform further analysis. Therefore, the resume parsing module acts as the primary data provider for the entire recruitment system.

4.5 Data Flow Explanation

The system follows a sequential data processing pipeline. Candidate resumes are first uploaded and converted into machine-readable text. The text is then processed using NLP techniques, and a CRF-based NER model extracts key entities. These entities are analyzed to estimate skill proficiency using contextual information, and the resulting candidate profile is stored in MongoDB. The structured data is then used by other system modules to generate recruitment insights.

4.6 Scalability and Security Considerations

The architecture is designed to be scalable so that large numbers of resumes can be processed efficiently. The modular design allows each component to be improved independently without affecting the overall system. The use of MongoDB enables flexible storage of semi-structured resume data and supports high-volume recruitment environments.

To ensure security, sensitive candidate information will be handled carefully and access will be restricted to authorized users. Personal data will be protected or anonymized when necessary to maintain privacy and comply with ethical data handling practices.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Several stakeholders interact with or benefit from the proposed intelligent recruitment system. These stakeholders include recruiters, job applicants, and system administrators who rely on the system to support recruitment activities.

Recruiters / Hiring Managers are the primary users of the system. They use the platform to analyze candidate profiles, review extracted skills and experience, and make informed hiring decisions using the insights generated by the system.

Job Applicants interact with the system by submitting their resumes and participating in recruitment processes. The system analyzes their resumes and generates structured candidate profiles used during evaluations.

System Administrators manage the technical operation of the system. They are responsible for maintaining the system infrastructure, managing data storage, monitoring system performance, and ensuring the system operates securely and efficiently.

5.2 Functional Requirements

1. The system must provide the following functional capabilities:
2. The system shall allow users to upload candidate resumes in formats such as PDF or DOCX.
3. The system shall extract textual information from uploaded resumes using document processing tools.
4. The system shall preprocess resume text using Natural Language Processing (NLP) techniques such as tokenization and text normalization.
5. The system shall identify entities such as skills, education, work experience, and project information using a Named Entity Recognition (NER) model.
6. The system shall analyze contextual information such as experience duration, project descriptions, and job roles to estimate skill proficiency.
7. The system shall generate a structured candidate skill profile based on the extracted information.
8. The system shall store candidate profiles and extracted data in a MongoDB database.
9. The system shall provide structured candidate information to other recruitment modules such as job–candidate matching and recruitment analytics.

5.3 Non-Functional Requirements

In addition to functional capabilities, the system must satisfy several non-functional requirements to ensure reliability, performance, and security.

Security:

The system must ensure that candidate data is securely stored and accessed only by authorized users. Sensitive information such as personal details should be protected or anonymized to maintain data privacy.

Performance:

The system should process resume documents efficiently and provide results within a reasonable time frame, even when handling multiple resume submissions.

Scalability:

The system should be capable of processing large volumes of resumes without performance degradation, supporting recruitment environments with many applicants.

Usability:

The system should provide a user-friendly interface that allows recruiters to easily upload resumes, view candidate profiles, and access recruitment insights.

Reliability:

The system should operate consistently and accurately during resume processing and skill analysis, minimizing errors in information extraction and candidate evaluation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed system targets organizations that handle large volumes of recruitment data and require efficient candidate screening processes. The primary target market includes corporate human resource (HR) departments, recruitment agencies, and technology companies that regularly process large numbers of job applications.

The main customer persona is the HR Manager or Recruitment Specialist, who is responsible for screening candidate resumes and selecting suitable applicants. These professionals require tools that can quickly analyze resumes, identify candidate skills, and support data-driven hiring decisions. The system can also benefit large enterprises and startups that want to improve recruitment efficiency through AI-driven automation.

6.2 Value Proposition

The proposed system provides an AI-powered recruitment intelligence platform that automates resume analysis and enhances hiring decisions. Unlike traditional Applicant Tracking Systems (ATS) that rely mainly on keyword matching, the system introduces context-aware skill proficiency analysis. This allows recruiters to better understand the depth and relevance of candidate skills by analyzing contextual information such as experience duration and project involvement.

By transforming unstructured resume data into structured candidate profiles, the system helps organizations reduce manual screening time, improve hiring accuracy, and identify qualified candidates more effectively.

6.3 Revenue Model

The system can be commercialized through a Software-as-a-Service (SaaS) model. Organizations can subscribe to the platform and access its recruitment analysis features through a web-based interface. The platform may offer different subscription tiers depending on the number of users, recruitment volume, and additional analytical features.

Another possible revenue source is enterprise licensing, where large organizations purchase customized deployments of the system integrated into their existing HR platforms.

6.4 Cost Estimation

The primary costs involved in developing and operating the system include software development, cloud infrastructure, and system maintenance. Development costs include implementing NLP models, system architecture design, and integration of AI components. Operational costs include server hosting, database management (MongoDB), and system monitoring.

Additional costs may arise from model training, data storage, and system upgrades, particularly when scaling the system to handle large recruitment datasets.6.5 Pricing Strategy

The pricing strategy will follow a subscription-based model where organizations pay a monthly or annual fee based on their usage. For example, smaller organizations may subscribe to a basic plan with limited resume processing capacity, while larger enterprises may select premium plans that support higher processing volumes and advanced analytics features.

This tiered pricing strategy allows the platform to be accessible to both small businesses and large organizations.

6.6 Competitive Advantage

The main competitive advantage of the proposed system lies in its context-aware skill proficiency estimation capability. Most existing recruitment systems rely primarily on keyword-based resume screening, which often fails to evaluate the actual depth of candidate skills.

The proposed approach uses NLP-based resume parsing and contextual skill analysis to estimate the relevance and expertise level of candidate skills. This enables more accurate candidate evaluation and provides recruiters with deeper insights into candidate capabilities, improving the overall recruitment process.

6.7 Intellectual Property Considerations

The intellectual property of the system primarily involves the algorithmic design and implementation of the context-aware skill proficiency analysis module. The software architecture, trained models, and system implementation can be protected through software copyrights and licensing agreements.

In addition, organizations deploying the system will retain ownership of their recruitment data, while the system developers maintain ownership of the underlying AI models and software framework used to analyze resume data.

7. BUDGET AND JUSTIFICATION

7.1 Software Licenses

Most of the technologies used in the proposed system are open-source, which significantly reduces software licensing costs. The system will be developed using widely adopted open-source technologies such as Python, SpaCy, sklearn-crfsuite, and MongoDB.

Table 1.Software Tools and License Cost Estimation

Software	Estimated Cost	Justification
Python Programming Environment	Free	Open-source programming language used for system development
SpaCy NLP Library	Free	Used for NLP preprocessing and linguistic analysis
sklearn-crfsuite (CRF-NER)	Free	Used to implement the CRF-based Named Entity Recognition model
MongoDB Community Edition	Free	Used to store structured candidate profiles and resume data
Development Tools (VS Code / IDE)	Free	Used for coding, debugging, and system development

7.2 Cloud Services

Cloud services may be required for system deployment, database hosting, and storage of resume data. These services allow the system to operate in a scalable environment and support multiple users.

Table 2.Cloud Services Cost Estimation

Service	Estimated Cost	Justification
Cloud Hosting (AWS / Google Cloud / Azure)	LKR 15,000	Used to deploy and run the AI-based recruitment system
Cloud Storage	LKR 5,000	Used for storing resume files and processed data
MongoDB Atlas (Database Hosting)	LKR 5,000	Managed cloud database service for scalable data storage

7.3 Human Effort Cost

Although the research is conducted by students as part of an academic project, significant time and effort are invested in designing, implementing, and evaluating the proposed system. The estimated cost reflects the opportunity cost of student research time, as the time spent on this project could otherwise be used for professional employment.

Table 3.Human Effort Cost Estimation

Role	Estimated Cost	Justification
Student Researcher / Developer	LKR 150,000	Represents the time and effort spent on system design, development, testing, and research activities

7.4 Total Estimated Budget

The estimated total budget for the proposed research project is summarized below.

Table 4.Total Estimated Project Budget

Category	Estimated Cost
Hardware costs	LKR 0
Software Licenses	LKR 0
Cloud Services	LKR 25,000
Human Effort Cost	LKR 150,000
Total Estimated Budget	**LKR 175,000**

7.5 Budget Justification

The proposed budget focuses primarily on cloud infrastructure and research effort, as most development tools used in this project are open-source and do not require licensing fees. Cloud services are required to host the system and store candidate data, enabling scalable processing of resume documents. The inclusion of human effort cost reflects the significant time and expertise required for research, system development, model implementation, and evaluation of the proposed NLP-based resume parsing and skill proficiency analysis module**.**

8. WORK BREAKDOWN STRUCTURE (WBS)

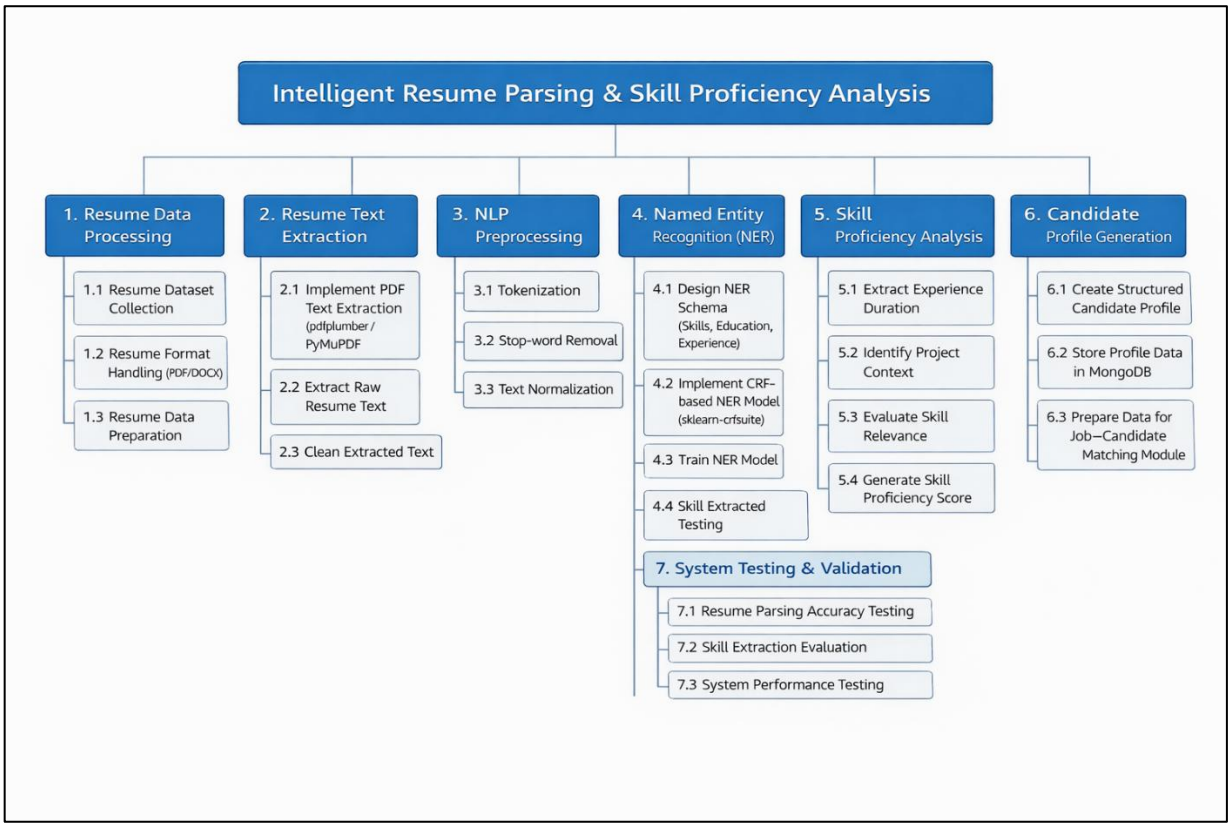


Fig. 5.Work Breakdown Structure

Figure 5 illustrates the Work Breakdown Structure (WBS) for the *Intelligent Resume Parsing & Skill Proficiency Analysis* module. The diagram presents the hierarchical decomposition of tasks required to implement the module, starting from resume data processing and text extraction to NLP preprocessing and Named Entity Recognition (NER). The extracted information is then used for context-aware skill proficiency analysis, followed by candidate profile generation and database storage. Finally, the system undergoes testing and validation to evaluate resume parsing accuracy, skill extraction performance, and overall system reliability. This structured breakdown helps organize the development process and ensures systematic implementation of the module.

9. GANTT CHART

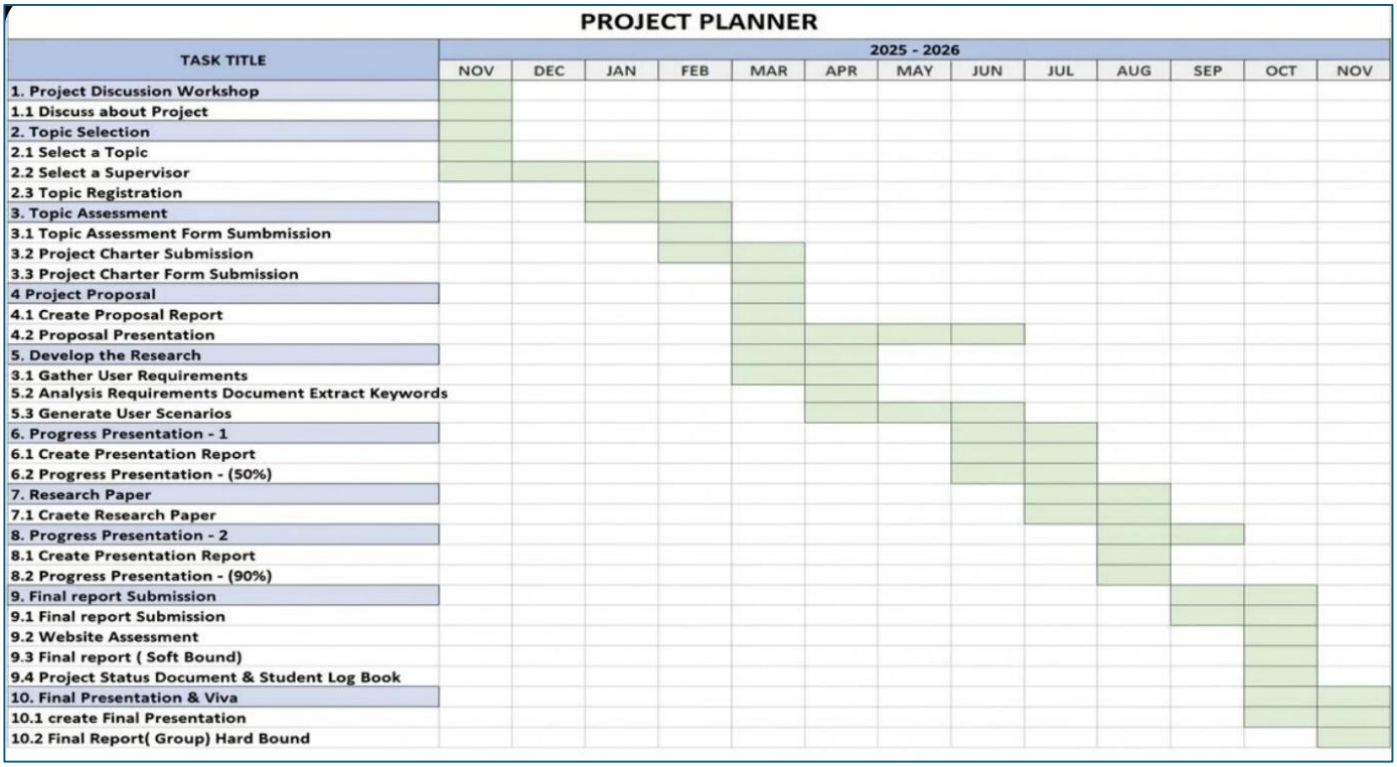


Fig. 6.Gantt Chart

10. REFERENCES LIST

[1] J. Smith and R. Brown, “Automated Resume Screening Using Natural Language Processing,” *Journal of Artificial Intelligence Research*, vol. 45, pp. 120–135, 2021.

[2] M. Johnson, “Challenges in Modern Recruitment and Talent Acquisition,” *International Journal of Human Resource Management*, vol. 30, no. 5, pp. 780–795, 2020.

[3] L. Rivera, “Hiring as Cultural Matching: The Case of Elite Professional Service Firms,” *American Sociological Review*, vol. 77, no. 6, pp. 999–1022, 2019.

[4] S. Gupta and P. Kumar, “Resume Information Extraction Using NLP Techniques,” *IEEE Access*, vol. 8, pp. 123456–123468, 2020.

[5] United Nations, “Sustainable Development Goal 8: Decent Work and Economic Growth,” United Nations, 2023.

[6] M. A. I. Mohammad Fadzir and S. Hashim, “SmartResume: A Resume Parser Application Using Natural Language Processing,” *Journal of Human Centered Technology*, vol. 5, no. 1, pp. 15–20, 2026.

[7] A. M. Al-Aswad, H. M. Al-Qamati, and H. S. Abu Alhaji, “An Intelligent Resume Analysis and Job Recommendation System Using NLP,” *Comprehensive Journal of Science*, vol. 10, no. 38, pp. 2615–2625, 2025.

[8] K. P., V. Prabhu, S. G., and D. De Hilster, “Automated Resume Parsing and Candidate Profiling System for Efficient Talent Acquisition,” in *Proc. International Conference on Pervasive Computing and Social Networking (ICPCSN)*, IEEE, 2025.

[9] T. K., D. R. J., L. G., N. Ahirwar, and S. Gajendran, “Automated Resume Parsing and Ranking Using Natural Language Processing,” in *Proc. IEEE International Conference on Artificial Intelligence for Internet of Things (AIIoT)*, IEEE, 2024.

[10] C. V. Raja, P. G. Sekaran, and A. R. Patrick Bernard, “HireSmart AI: Intelligent Resume Parsing and Candidate Assessment Using Natural Language Processing,” *International Journal of Sciences and Innovation Engineering*, vol. 3, no. 2, pp. 592–601, 2026.

[11] A. Julian and K. Haripriya, “NLP Based Resume Analysis and Adaptive Skill Assessment System,” in *Proc. IEEE International Conference for Innovation in Technology (INOCON)*, IEEE, 2024.

[12] N. Gangoda, N. Induvara, K. P. Yasantha, S. Thelijjagoda, C. Sewwandi, and N. Giguruwa, “Resume Ranker: AI-Based Skill Analysis and Skill Matching System,” in *Proc. International Conference on Intelligent Computing in Data Sciences (ICDS)*, IEEE, 2024.

[13] R. Deepa, V. Jayalakshmi, K. Karpagalakshmi, S. M. Prabhu, and P. Thilakavathy, “Survey on Resume Parsing Models for JOBCONNECT+: Enhancing Recruitment Efficiency Using Natural Language Processing and Machine Learning.

11. APPENDICES

Appendix A:Survey Instruments

To better understand the challenges faced by recruiters in resume screening and candidate evaluation, a survey instrument is proposed for Human Resource (HR) professionals and recruitment specialists. The purpose of this survey is to collect insights about the limitations of existing recruitment tools and the need for intelligent resume analysis systems.

Sample Survey Questions:

1. How many resumes does your organization typically receive for a single job opening?
2. What challenges do you face when manually screening resumes?
3. Does your organization currently use an Applicant Tracking System (ATS)?
4. How effective are existing systems in identifying candidate skills and experience?
5. How important is it for recruitment tools to evaluate actual skill proficiency instead of keyword matching?
6. Would AI-based resume analysis improve the efficiency of recruitment processes in your organization?

The results of this survey will help validate the research problem and support the development of the proposed system.

Appendix B:Preliminary UI Mockups

- This appendix presents preliminary user interface mockups that demonstrate how recruiters may interact with the proposed system. These mockups are early conceptual designs intended to illustrate potential system functionality and usabilitycandidate skill profile.

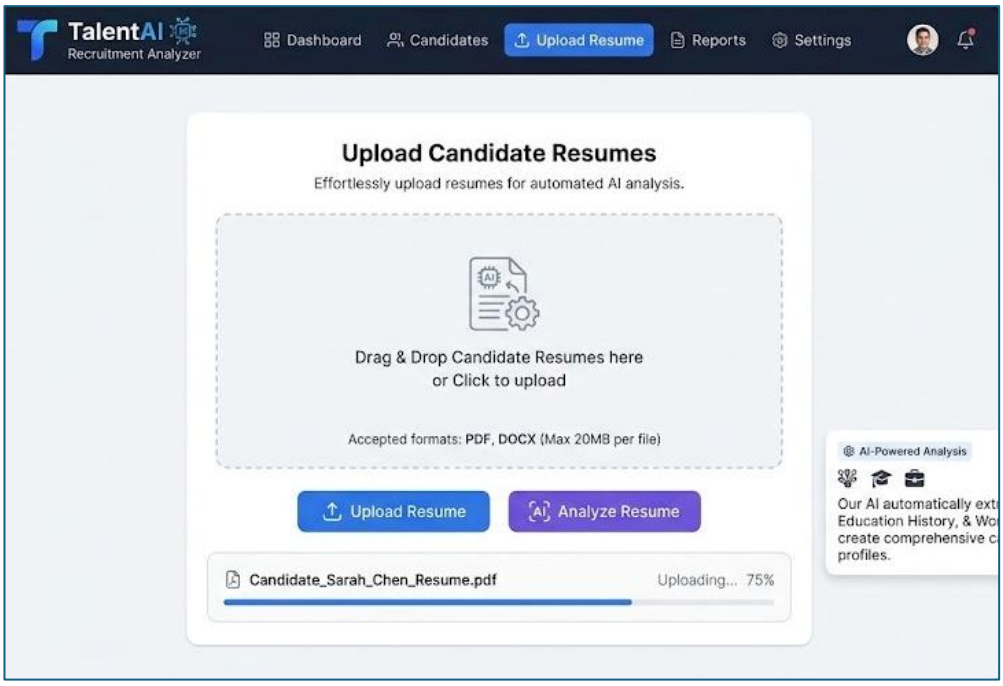


Fig. 7.Resume Upload Interface

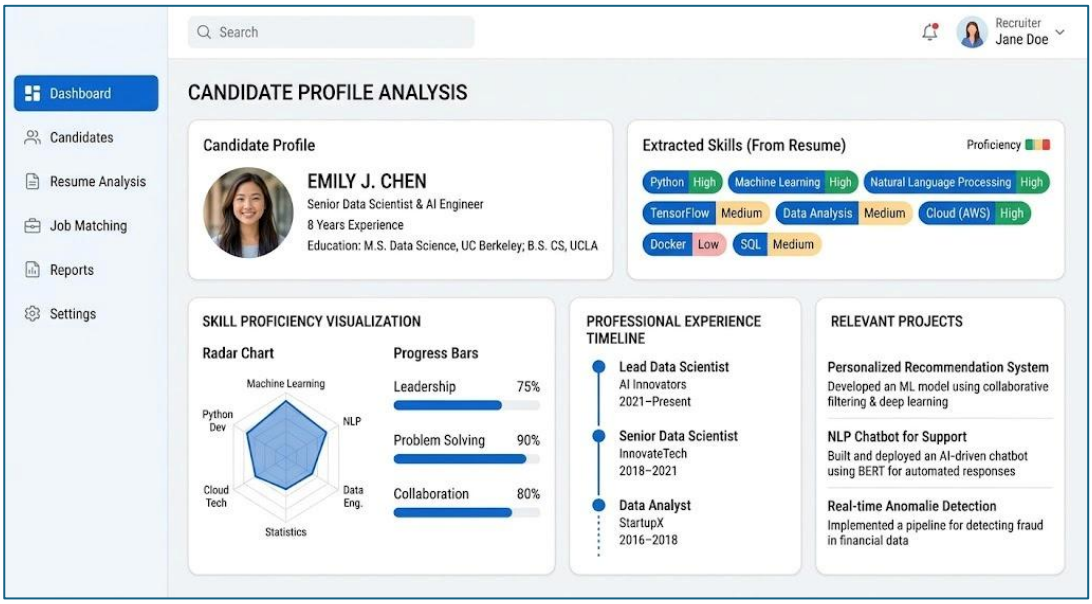


Fig. 8.Candidate Skill Profile Dashboard

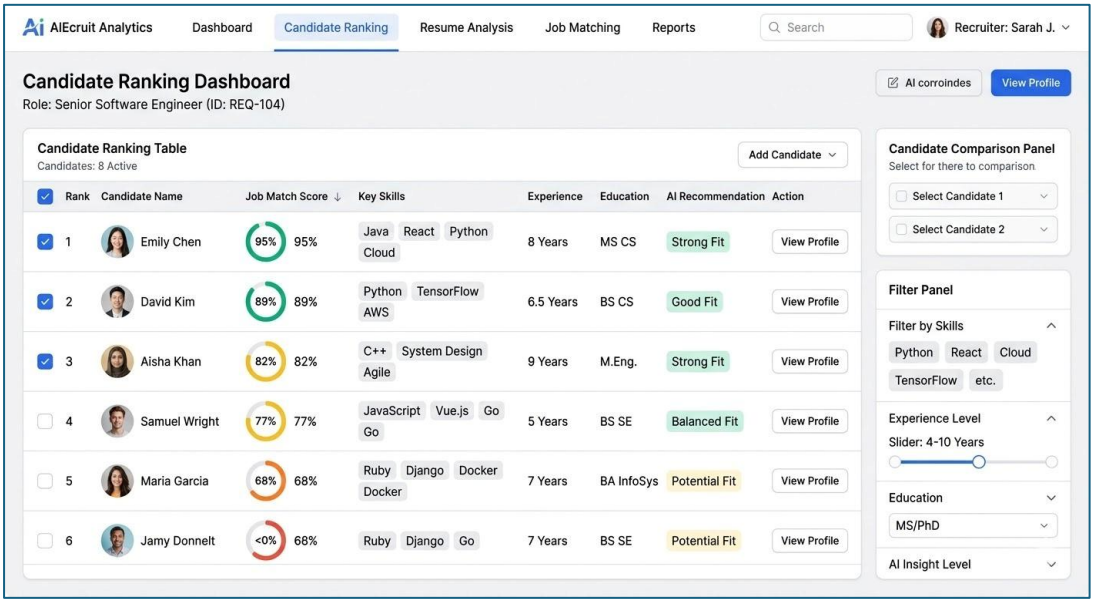


Fig. 9.Candidate Ranking & Analysis Page

Appendix C:Risk Analysis

The development of the proposed system involves several potential risks that may affect system performance, accuracy, and data security. The following table summarizes the identified risks and mitigation strategies.

Table 5.Risk Analysis and Mitigation Strategies

Risk ID	Risk Description	Impact	Mitigation Strategy
R1	Inconsistent resume formatting	May affect text extraction accuracy	Use robust extraction tools such as pdfplumber or PyMuPDF
R2	Limited training data for NLP models	Reduced accuracy in entity recognition	Use publicly available datasets and manual annotation
R3	Incorrect skill extraction	May affect skill proficiency estimation	Implement contextual validation techniques
R4	Large volume of resume data	May reduce system performance	Use scalable database solutions such as MongoDB
R5	Data privacy concerns	Risk of exposing candidate information	Implement data anonymization and access control mechanisms

Appropriate mitigation strategies will be applied to minimize these risks during system development and evaluation.

Appendix D:Ethical Documents

Ethical considerations are important when handling candidate data and personal information. This research follows ethical guidelines to ensure responsible data collection and processing.

Key ethical considerations include:

- Data Privacy Protection – Personal identifiable information such as names, email addresses, and contact details will be anonymized before analysis.
- Consent and Responsible Data Use – Resume data used in the research will be obtained from publicly available datasets or anonymized sources.
- Secure Data Storage – Candidate data will be securely stored and accessed only by authorized users.
- Research Integrity – The research will ensure transparency, accuracy, and responsible use of AI technologies in recruitment analysis.

These ethical guidelines ensure that the proposed research complies with responsible data usage and research ethics standards.